Luigi Borda

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PhD candidate in Mechanical Engineering specializing in neuroengineering and machine learning for human motor control. Expertise in EMG signal processing, wearable sensing, and closed-loop neuromodulation, with a strong focus on real-world deployment. Experience building data-driven systems for motor recovery assessment and assistive human-machine interaction.

Education

Carnegie Mellon University, Pittsburgh, PA
PhD in Mechanical Engineering — **GPA: 3.87/4.00 -** Expected Defense December 2026
Focus: Neuroengineering, machine learning for motor control, wearable sensing

Polytechnic University of Turin, Italy
Master of Science in Biomedical Engineering — **GPA: 110/110 (cum laude) -** Mar 2021
Bachelor of Science in Biomedical Engineering — **GPA: 107/110 -** Jul 2018

Skills

Research & Domain Expertise: Motor control, machine learning, electromyography (EMG), neuromechanics, sensory feedback, neuromodulation

Machine Learning & Signal Processing: Time-series analysis, neural decoding, deep learning, feature extraction, statistical modeling

Programming Languages: Python, Matlab, C++, C#, Assembly

Libraries & Frameworks: PyTorch, NumPy, SciPy, scikit-learn

Engineering Tools & Platforms: Azure DevOps, Git/GitKraken, Vicon Nexus, Unity

Languages: Italian (native), English, Spanish

PhD Research

NeuroMechatronics Lab (Prof. Douglas J. Weber), Mechanical Engineering Dept., CMU AUG 2022 - DEC 2026

Gesture decoding for brain–computer interface (BCI) applications using wearable sEMG in post-paralysis subjects

- Developed a real-time multimodal (EMG–IMU) gesture decoding pipeline using Meta's Neural Band enabling intuitive hand gesture control in individuals with paralysis
- Designed noise-robust feature extraction methods for reliable motor intent decoding under wearable sensing constraints
- Validated closed-loop performance in human subjects, demonstrating stable real-time interaction with digital interfaces
- Collaborated with Meta Reality Labs on system development and evaluation, delivering milestone reports, performance benchmarks, and iterative model improvements aligned with project objectives

At-home monitoring of motor recovery post-stroke using wearable sensing

- Developed a remote wearable sensing pipeline using Empatica Embrace Plus to track longitudinal motor recovery in stroke patients outside clinical settings
- Validated wearable-derived biomarkers against clinical motor assessments to establish their relevance to motor recovery
- Modeled longitudinal recovery trajectories from real-world wearable data, enabling objective tracking of motor function

AI-based optimization of spinal cord stimulation for rehabilitation in people with post-stroke motor deficits

- Developed Gaussian Process-based Bayesian optimization framework to optimize neuromodulation parameters
- Integrated EMG and kinematic biomarkers to define a physiologically informed objective function for optimization
- Assessed efficacy of closed-loop stimulation for motor rehabilitation in people with post-stroke motor deficits
- Contributed to IP development and ongoing collaboration with Reach Neuro Inc., with the system being integrated into a patent application for future neurotechnology products

Understanding and decoding neural activity in people with spinal muscular atrophy

- Designed and implemented a real-time neural detection framework supporting motor unit–based control interactions
- Developed experimental paradigms combining force sensing and high-density sEMG for motor assessment under pathological conditions
- Extracted and quantified neural properties using advanced signal processing methods for neuromuscular characterization

Work Experience

Medela AG, Baar, Switzerland SEPT 2021 - JUL 2022

Product Development Engineer HW/SW

- Developed embedded control software for DC motor–driven fluid systems in medical devices
- Designed and tuned PID control algorithms to ensure stable and reliable fluid drainage under daily-use conditions
- Supported implementation and testing of motor control software for integration into medical device prototypes

Research Experience

Neuroengineering Lab, ETH, Zürich, Switzerland SEPT 2020 - MAR 2021

Sensory feedback integration to enable advanced interaction in virtual reality environments

- Integrated TENS-based feedback, significantly enhancing perception in VR environments
- Designed virtual reality environments to assess proportional sensory feedback integration

AI optimization of somatosensory stimulation for sensory feedback restoration after peripheral neuropathy

- Developed reinforcement learning-driven VR framework for automated calibration, reducing manual tuning effort in neuromodulation studies
- Assessed sensory perception differences in individuals with sensory loss after peripheral neuropathy compared to an able-bodied cohort

Patent

WO2025096820A1: Active implantable medical devices, bioelectronics, neuromodulation and associated methods.
Inventors: Marc Powell, Douglas Weber, Marco Capogrosso, **Luigi Borda**, Angelica Herrera, Charli Ann Hooper

Publications

- Sorensen E., **Borda L.**, et al., "Recovery of Dexterous Motor Control via Non-Monosynaptic Corticospinal Pathways.", medRxiv (2026)
- Despradel D., Max Murphy, **Borda L.**, et al., "Enabling Skilled Human-Computer Interaction After Paralysis via a Wearable sEMG Interface", medRxiv (2026)
- Refy O., **Borda L.**, Jacob Hsu, et al., "Spinal Cord Stimulation Improves Feedback Control of Reaching Post-Stroke.", medRxiv (2025)
- **Borda L.**, Verma N.,et al., "Spinal cord stimulation modulates post-synaptic inhibition to improve arm neuromotor control after chronic stroke.", Cell Reports Medicine (2026)
- Prat-Ortega G., Ensel S., Donadio S., **Borda L.**, et al., "First-in-human study of epidural spinal cord stimulation in individuals with spinal muscular atrophy", Nature Medicine (2025)
- Balaguer J., Prat-Ortega G., Ostrowski J., **Borda L.**,et al., "Neural mechanisms underlying the recovery of voluntary control of motoneurons after paralysis with spinal cord stimulation." Neuron (2025)
- **Borda L.**, Gozzi N.,et al., "Automated calibration of somatosensory stimulation using reinforcement learning", Journal of NeuroEngineering and Rehabilitation (2023)

Conferences

- *"Effects of Epidural Spinal Cord Stimulation on Posterior Root-Muscle Reflexes Evoked in Antagonistic Muscles"*, Presented at Society for Neuroscience, Meeting, 2024, Washington D.C., USA
- *"Underlying Changes in Motor Units Properties in Presence of Epidural Spinal Cord Stimulation in Individuals Affected by Spinal Muscular Atrophy"*, Presented at The International Society of Electrophysiology and Kinesiology, Congress, 2024, Nagoya, Japan
- *"Epidural spinal cord stimulation reduces agonist-antagonist co-contraction facilitating strength production"*, Presented at Society for Neuroscience, Meeting, 2023, San Diego, USA
- *"Epidural Spinal Cord Stimulation Inhibits Monosynaptic Reflexes in Antagonist Muscles"*, Presented at IEEE Engineering in Medicine and Biology Society, 2023, Sydney, Australia

Awards

- Dompé Foundation scholarship, 2025
- Most Entrepreneurial Pitch, BioHackathon Nucleate PGH, Pittsburgh, PA, 2024
- 2nd place in Health Hackathon, BioHackathon Nucleate PGH, Pittsburgh, PA, 2024
- 2nd place in Patient Safety Challenge, BioHackathon Nucleate PGH, Pittsburgh, PA, 2024
- PhD research symposium award, Pittsburgh, PA, 2023

Graduate Coursework

PhD courses: Machine Learning, Biomechanics of Human Movement, Statistical Models of the Brain, Engineering a Startup, Numerical Methods
Master courses: Neural Signal Processing, Image Processing, Design of Biomedical Programmable Systems